



Integrating satellite remote sensing and dynamic GIS for urban flood risk modelling in Ulaanbaatar,



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International Workshop on
Remote Sensing and **Geospatial Applications**



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RESEARCH OBJECTIVE

Alos Palsar satellite data to produce a flood risk map of Ulaanbaatar city ArcGis software, and to conduct a land use study of selected sites within the identified flood risk zones

OBJECTIVES

To study the current state of flood risk in Ulaanbaatar city

To comparatively study methods for determining flood risk

Downloading and Converting Satellite Data for Mongolian Conditions

To analyze spatial datasets and produce thematic maps

To produce a flood risk map of Ulaanbaatar city and identify at-risk districts

Land use study of selected areas located in flood risk zones

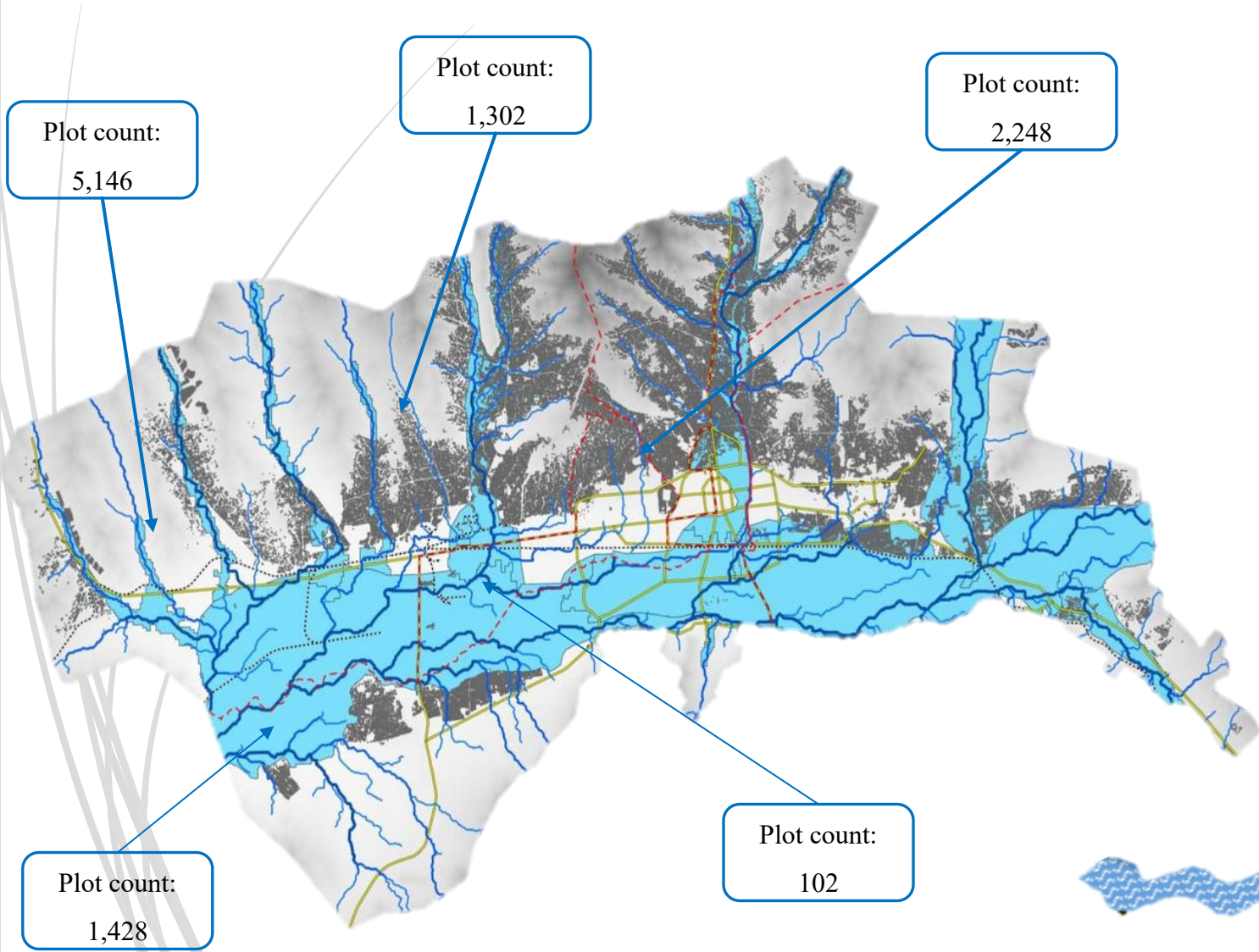


CURRENT STATUS OF FLOOD RISK IN ULAANBAATAR CITY

Residential zones prone to river flooding: 8

Residential zones prone to flash flooding: 3

Residential zones prone to mudflow flooding: 75



Flood Risk	District	Khoroo	Area (ha)	Share (%)
High Risk	Songinokhai rkhan	1,2, 3,4,5,7,8,9,10, 12, 13,14,15,16,17,20, 22, 23, 25, 26, 28, 29,30,31, 34, 35, 36, 38, 40,41,42	12,565 ha	29.8
	Chingeltei	7,11,12, 13,14,15,16	440 ha	1.04
	Sukhbaatar	13, 15,16,18,20, 1,7,8,10,11,12	2,315 ha	5.5
	Bayanzurkh	1,2,3,6,25,15,26,17,18	14,422 ha	34.2
	Bayangol	2,6,8,10,11,13,19,22,23, 20,25	1,105	2.6
	Khan-Uul	15	11,315	26.8



Downloading and Converting Satellite Data for Mongolian Conditions



ALOS PALSAR- digital elevation model at 12.5 m resolution

Precipitation data for 2011–2020 from “Version 4 of the CRU TS monthly high-resolution gridded multivariate climate dataset” was used.

2022 land cover data from “ESRI | Sentinel-2 Land Cover Explorer”

Landsat 8 satellite imagery at 30-meter resolution

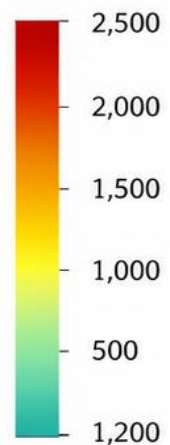


ULAANBAATAR

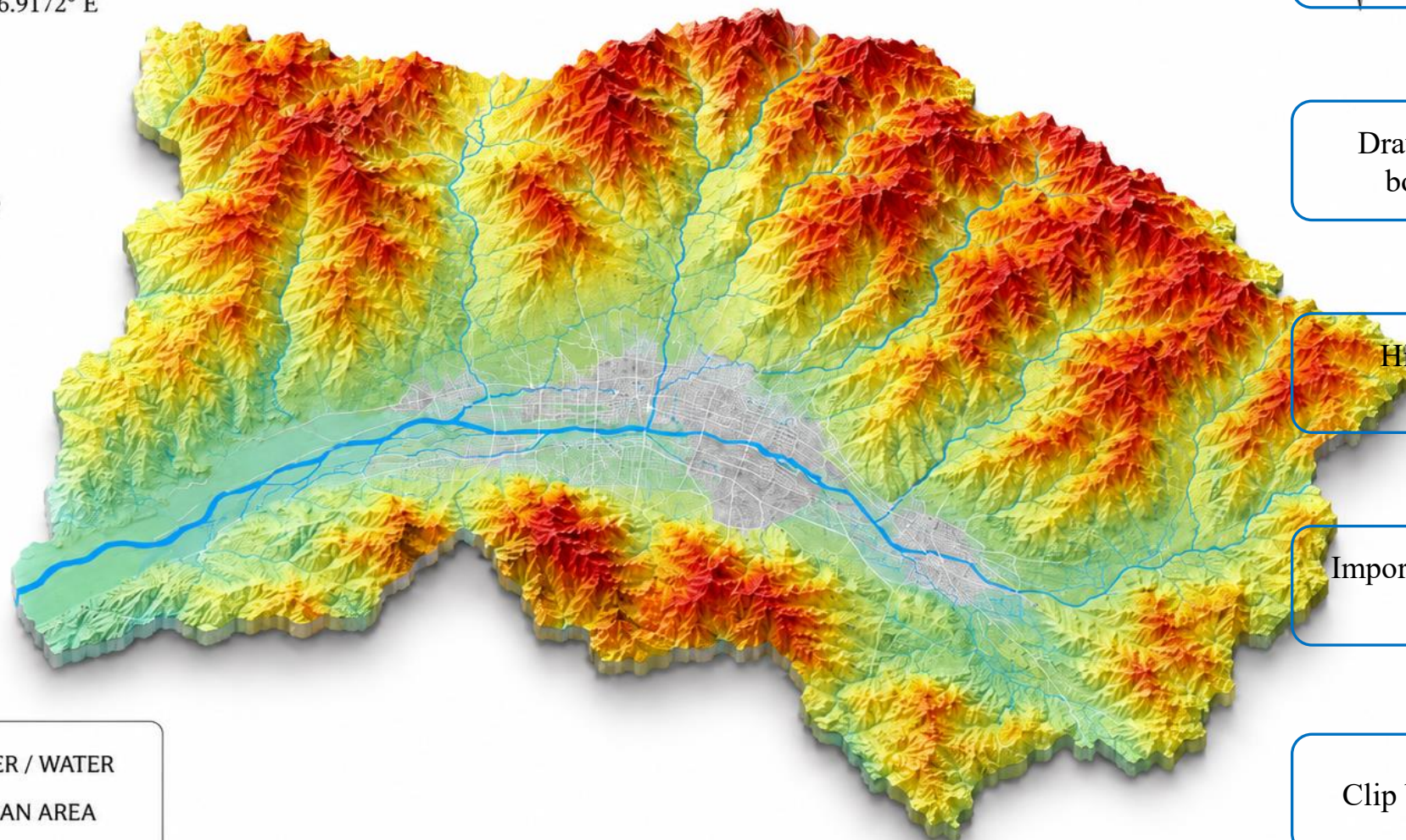
DIGITAL ELEVATION MODEL

47.9189° N, 106.9172° E

ELEVATION (m)



-  RIVER / WATER
-  URBAN AREA
-  ROADS



Draw and mark the study area boundary with a polygon.

High resolution (12.5) and download.

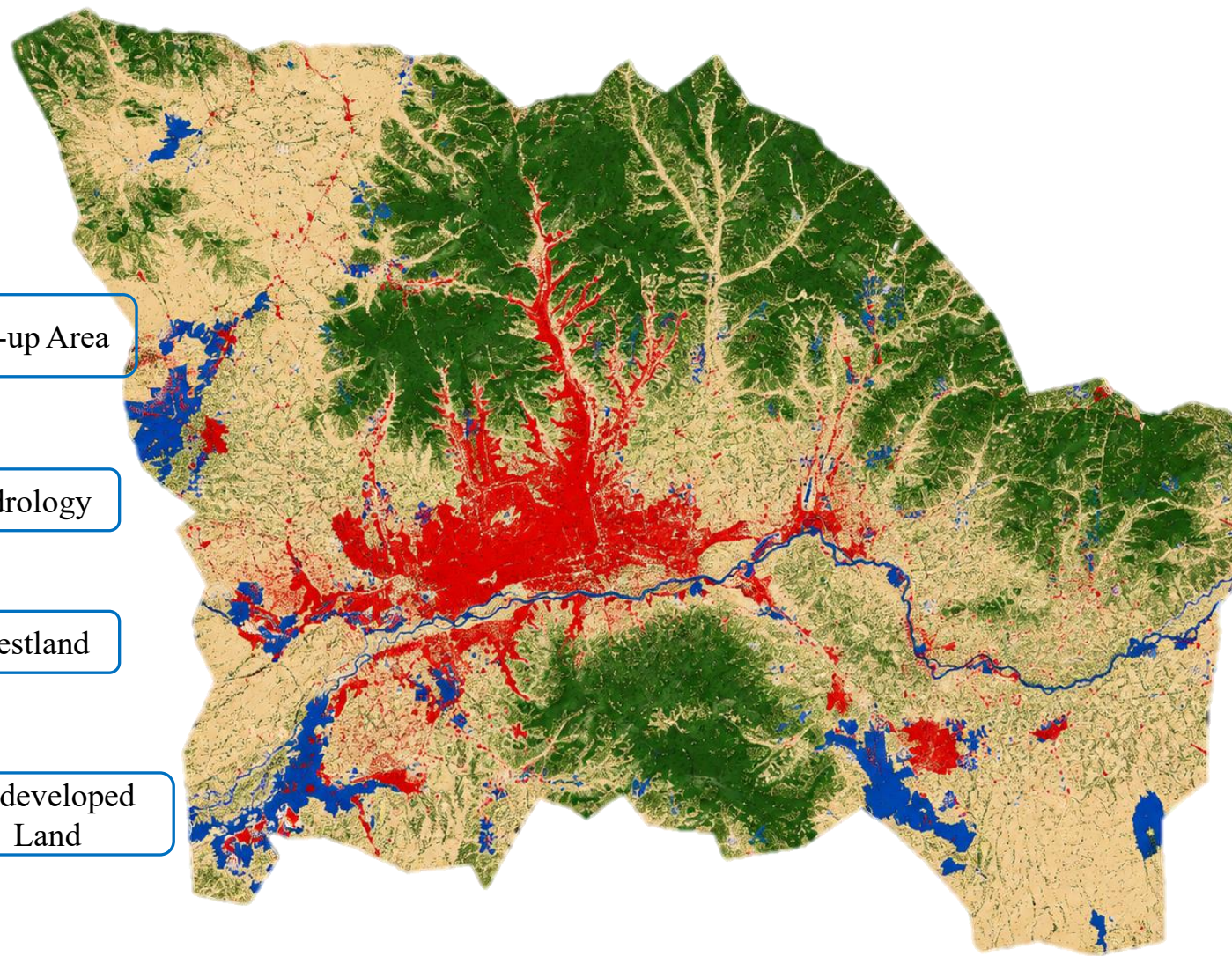
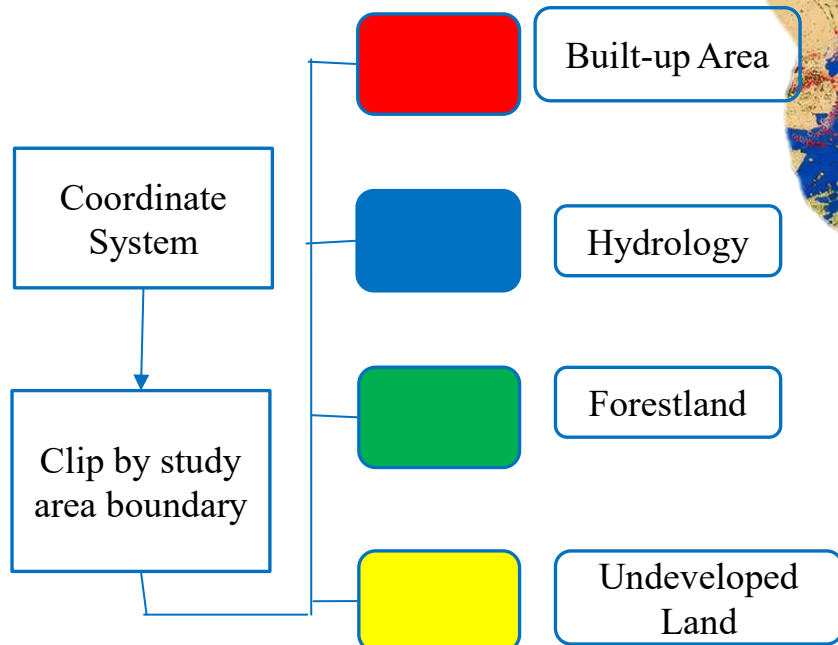
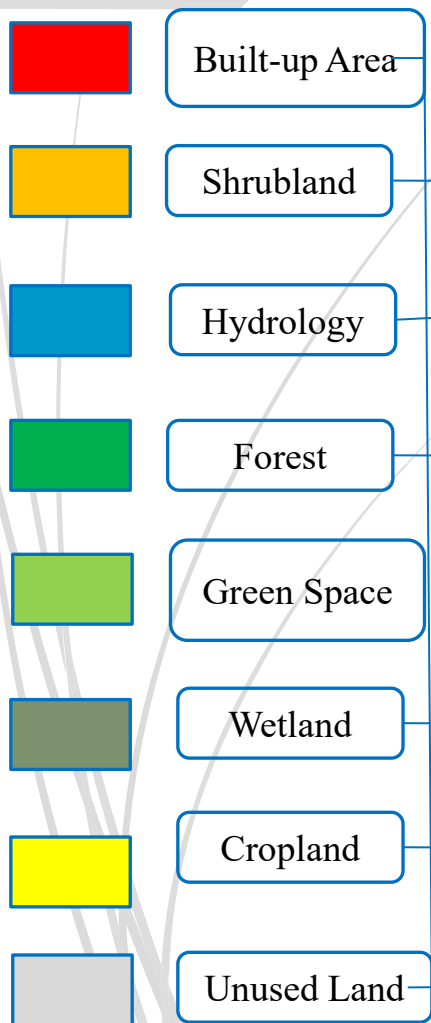
Import the downloaded DemDEM ArcGis into ArcGIS.

Clip by the study area boundary.

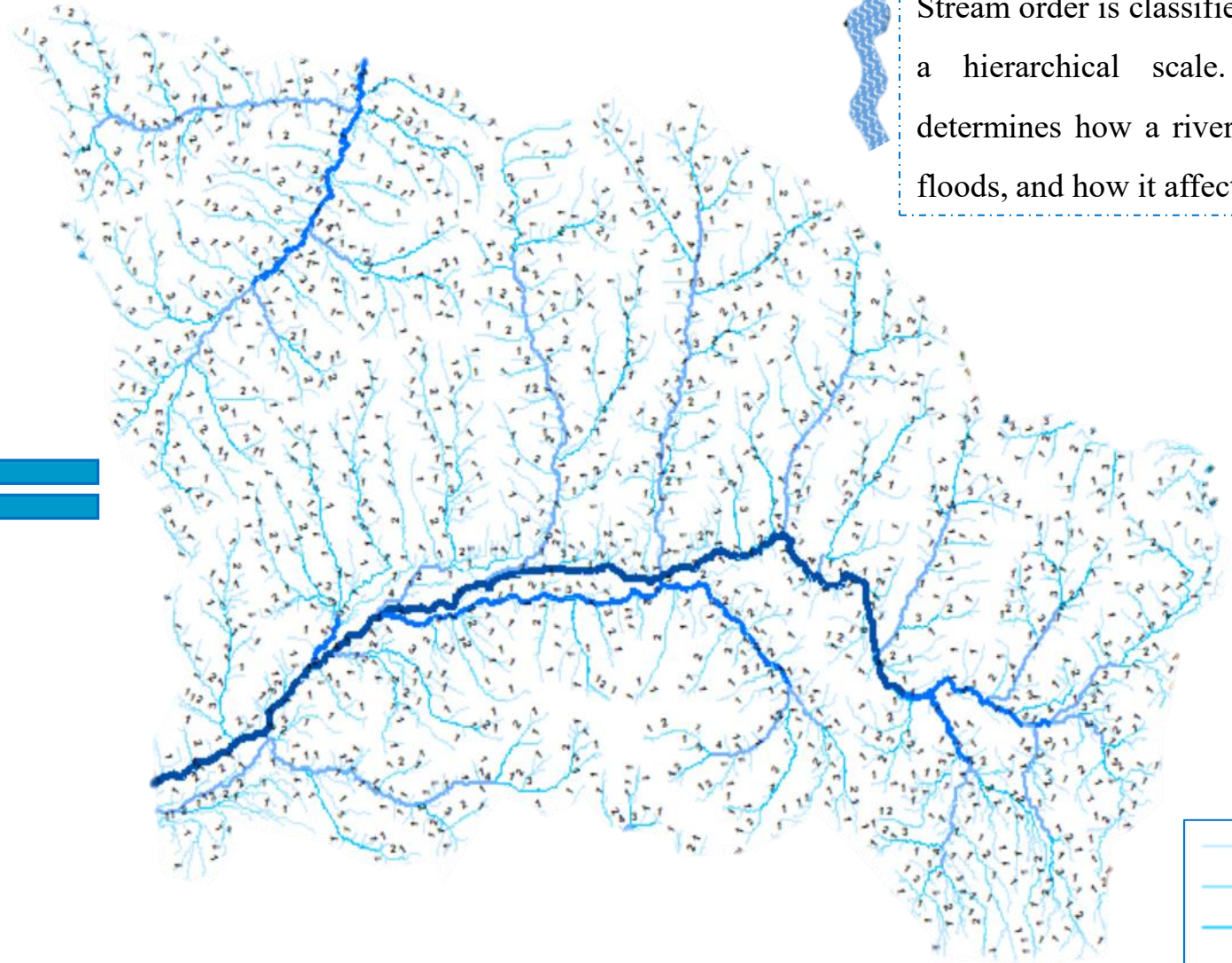
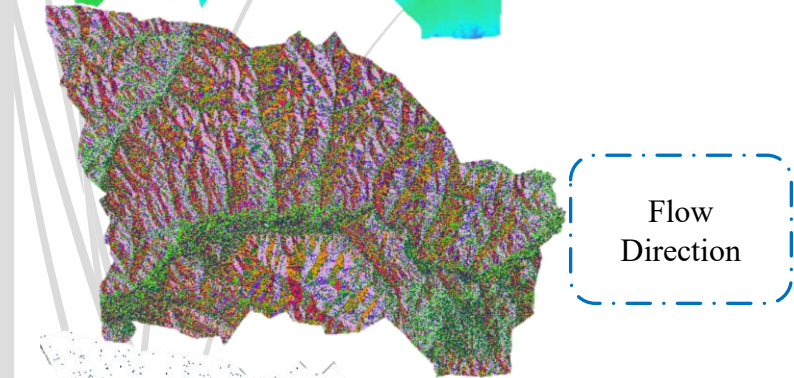
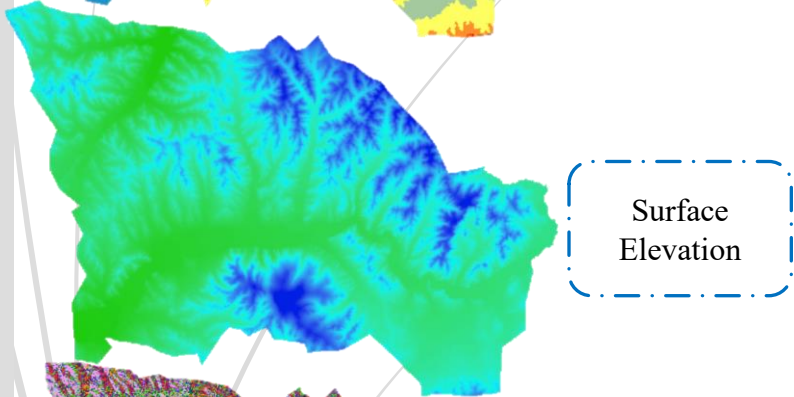
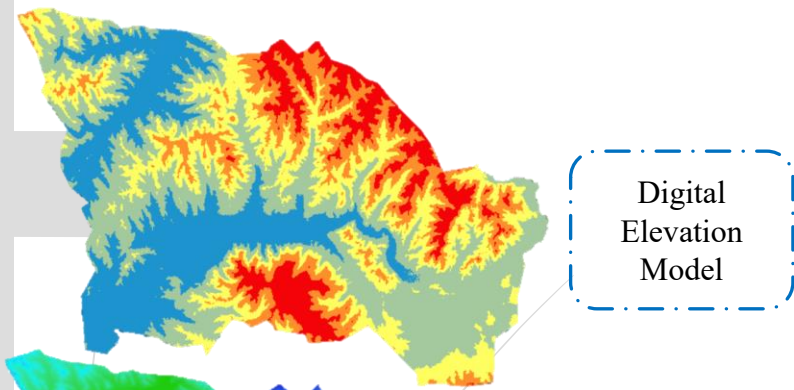




LAND COVER DATA



FLOW DIRECTION /HYDROLOGY/

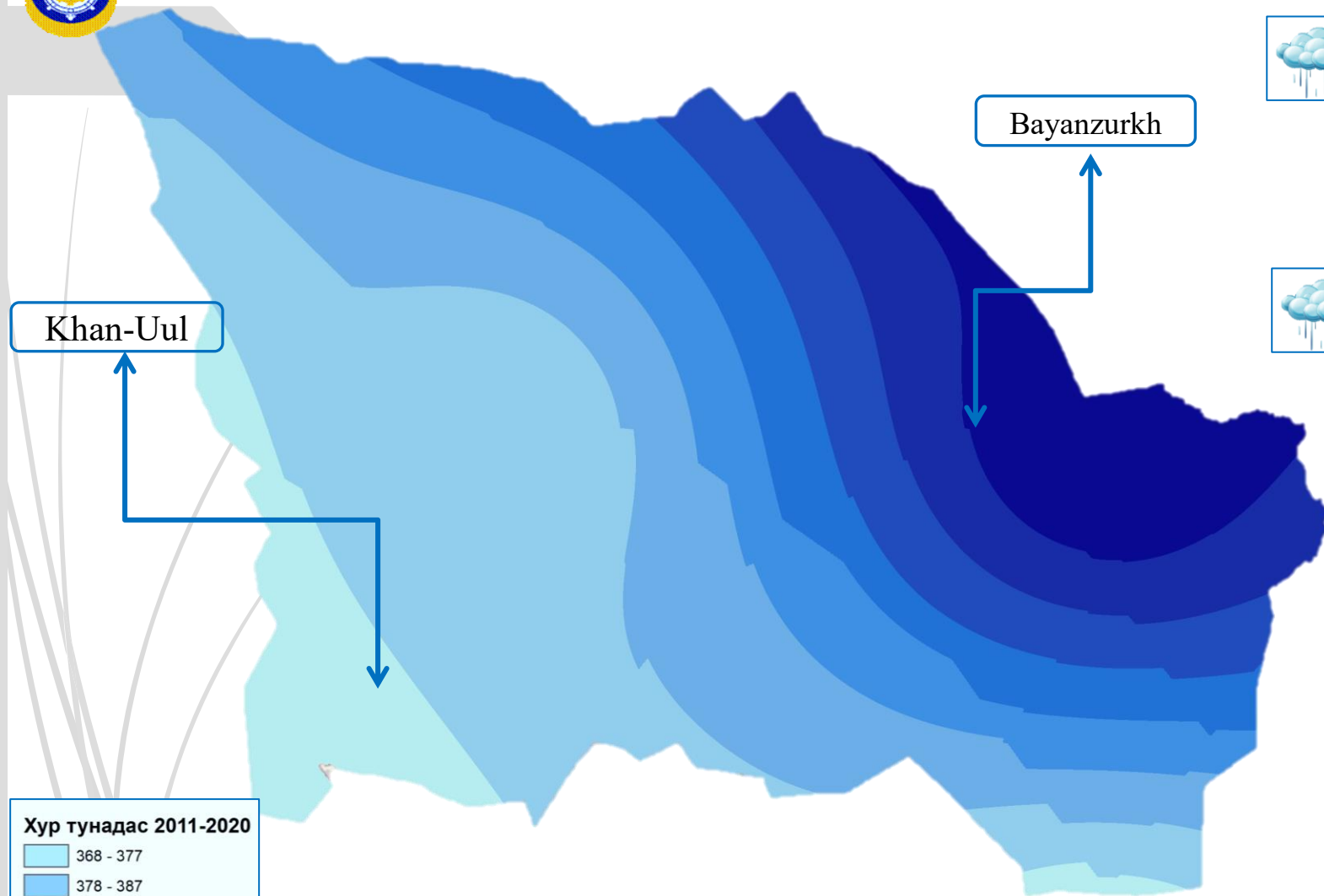


Stream order is classified from 1 to 6 on a hierarchical scale. Stream order determines how a river flows, where it floods, and how it affects water quality.

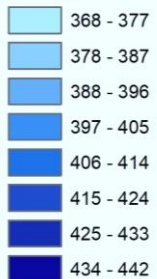




PRECIPITATION /INVERSE DISTANCE WEIGHTED/



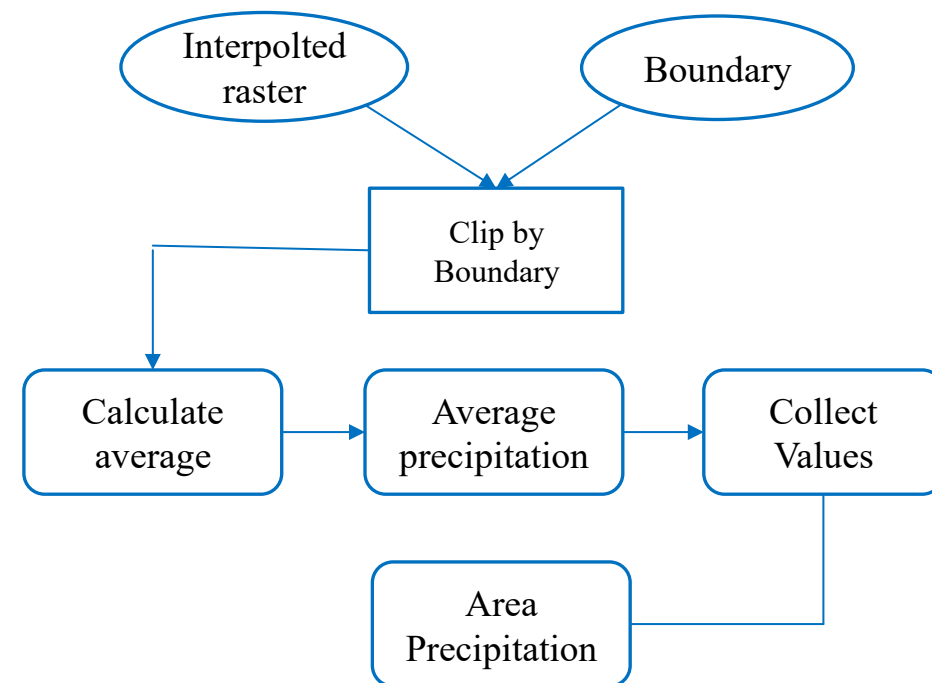
Хур тунадас 2011-2020



Precipitation data was determined using the Inverse Distance Weighted (IDW) interpolation method commonly used in hydrology.



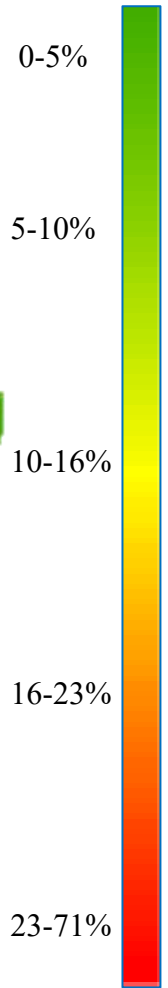
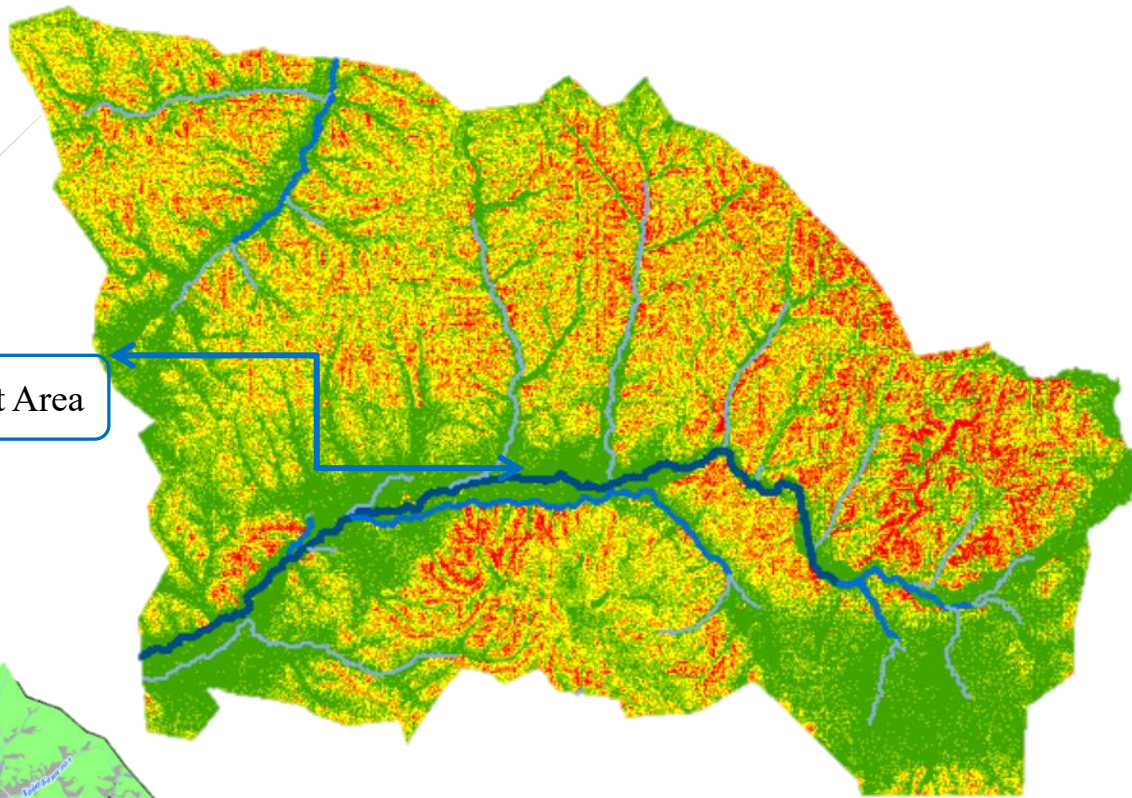
IDW- interpolated precipitation data can produce more accurate results for the summer season compared to other seasons.





Slope /Slope/

Central Settlement Area



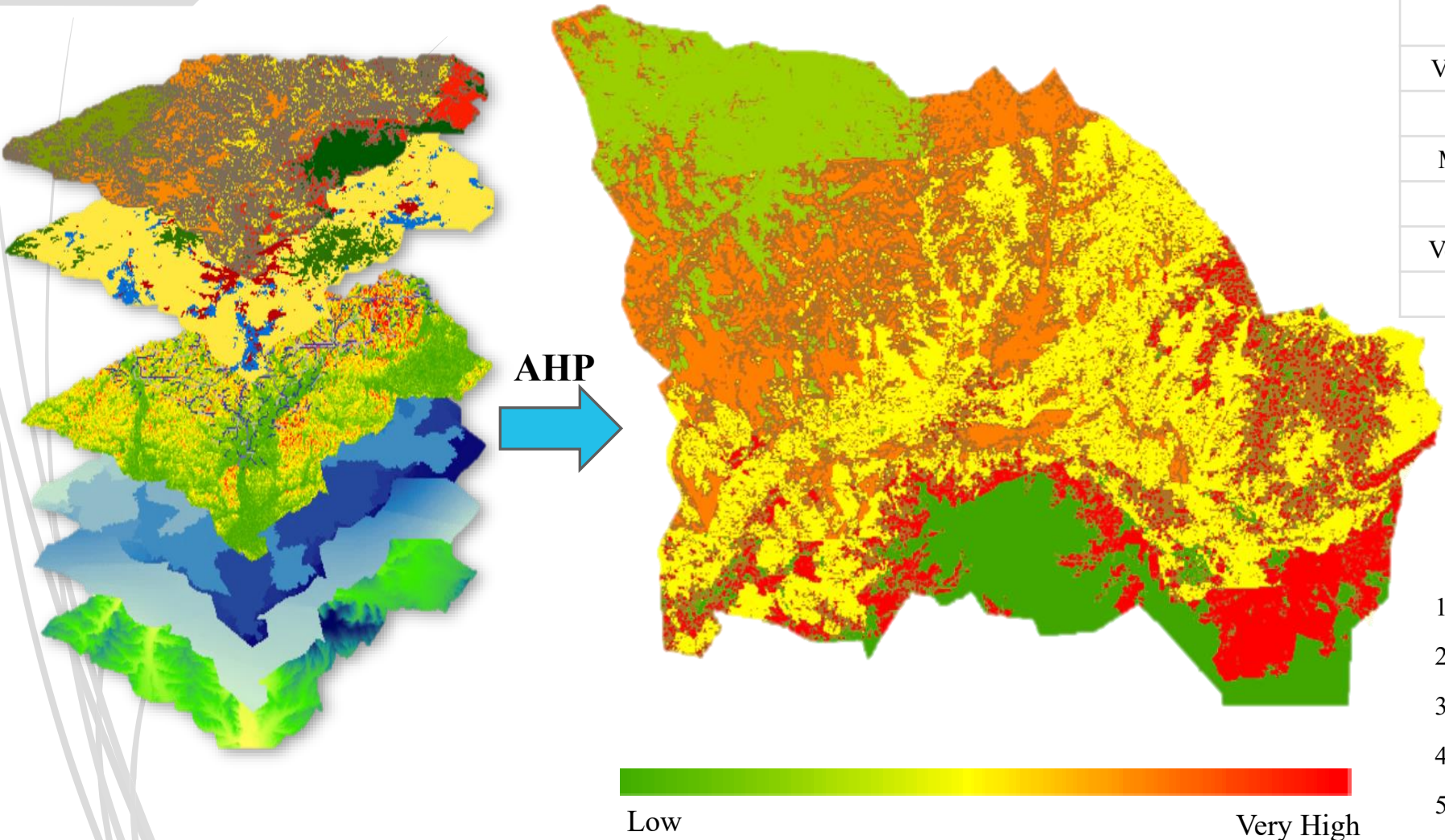
In steep terrain zones, water flows rapidly over the surface due to gravity, easily draining runoff to other areas.

Conversely, water tends to accumulate on flat terrain, which increases flood risk.

	Slope %	Area (km ²)	Share %
1	0-5	1295.98	32.78
2	5-10	1066.45	26.98
3	10-16	854.93	21.63
4	16-23	538.58	13.62
5	23-71	197.34	4.99



FLOOD RISK MAP OF ULAANBAATAR CITY (AHP)



Rating	Area (km²)	Share (%)
Very Low	446.7	11.3
Low	497.9	12.6
Medium	1218.8	30.9
High	1006.9	25.5
Very High	775.7	19.7
	3945.9911	100

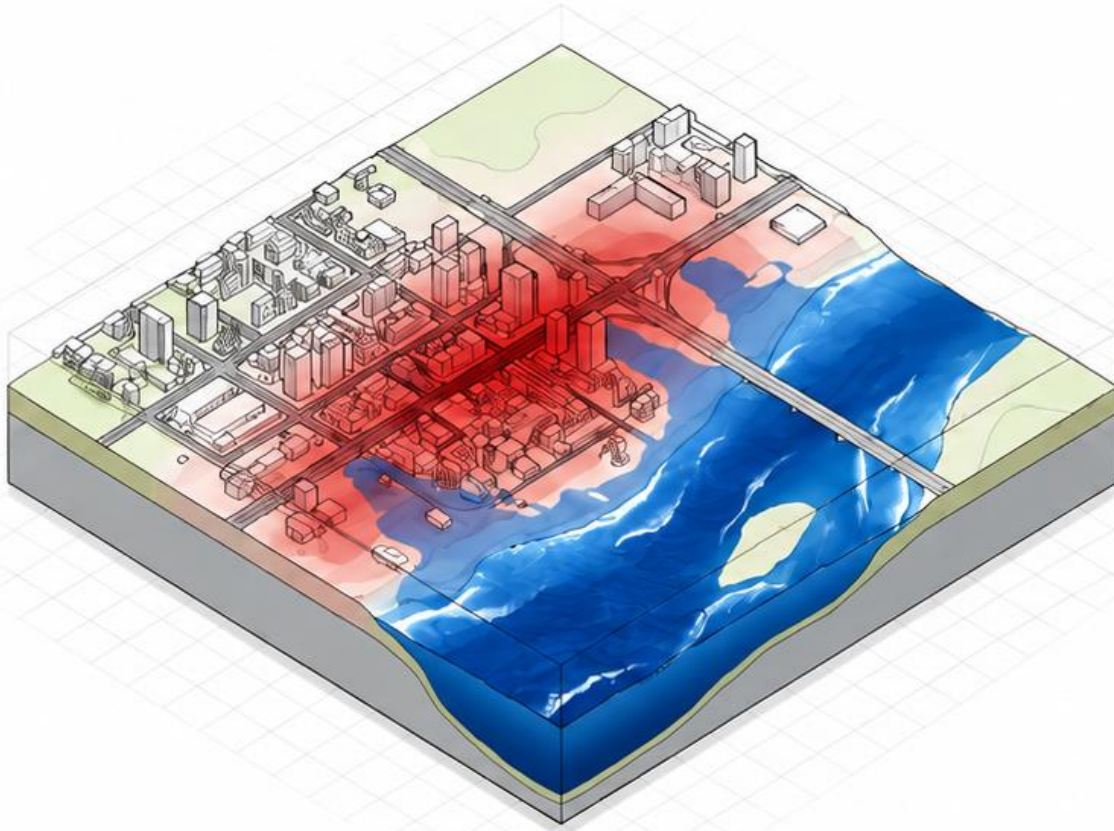
	Factor	Weight	Rank
1	Land Use	0.10	6
2	Digital Elevation Model	0.20	3
3	Flow Direction	0.25	1
4	Slope	0.21	2
5	Precipitation Amount	0.11	5
6	Watershed	0.13	4

$$P = W_{xSL} + W_{xFA} + W_{xD} + W_{xPr} + W_{xSO} + W_{xLU}$$



Riparian Zones

(The Tuul River Floodplain)

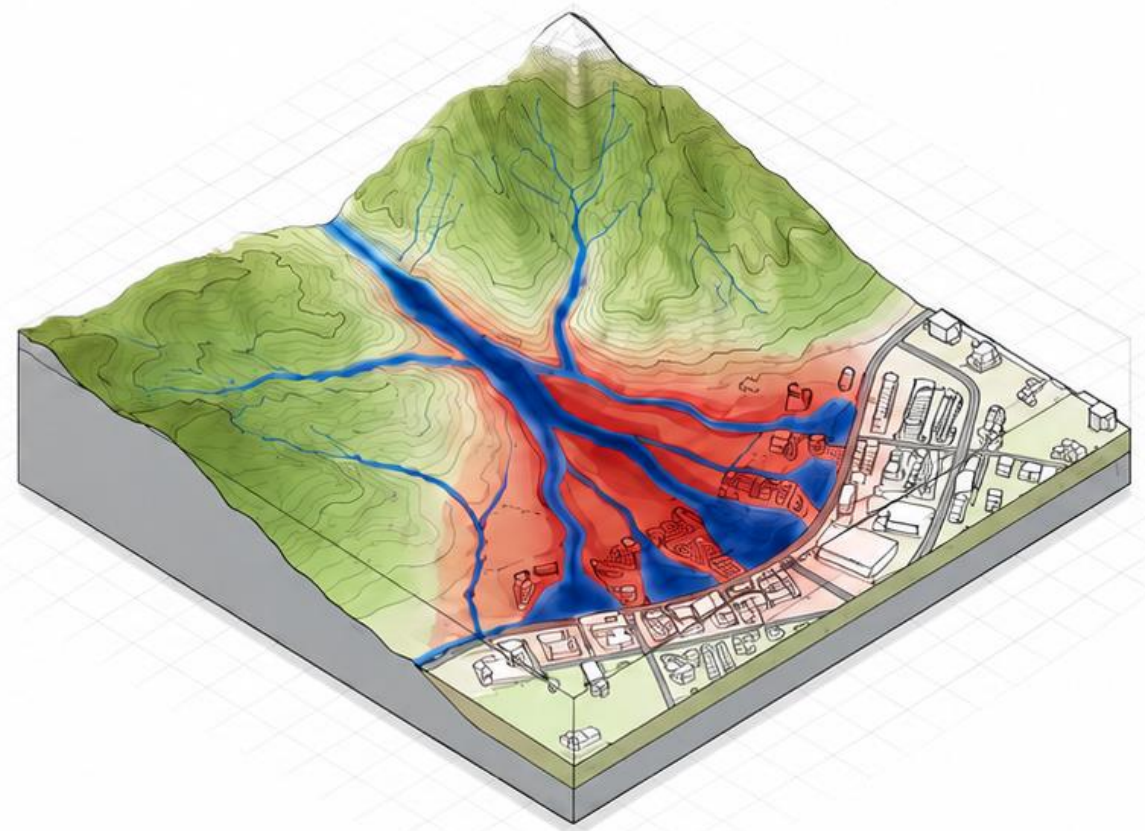


Mechanism: Hydrological forcing and riverine overflow potential.

Risk: Low-lying topography intersecting the primary watershed where the city's densest infrastructure is trapped.

Piedmont Areas

(Bogd Khan Foothills)

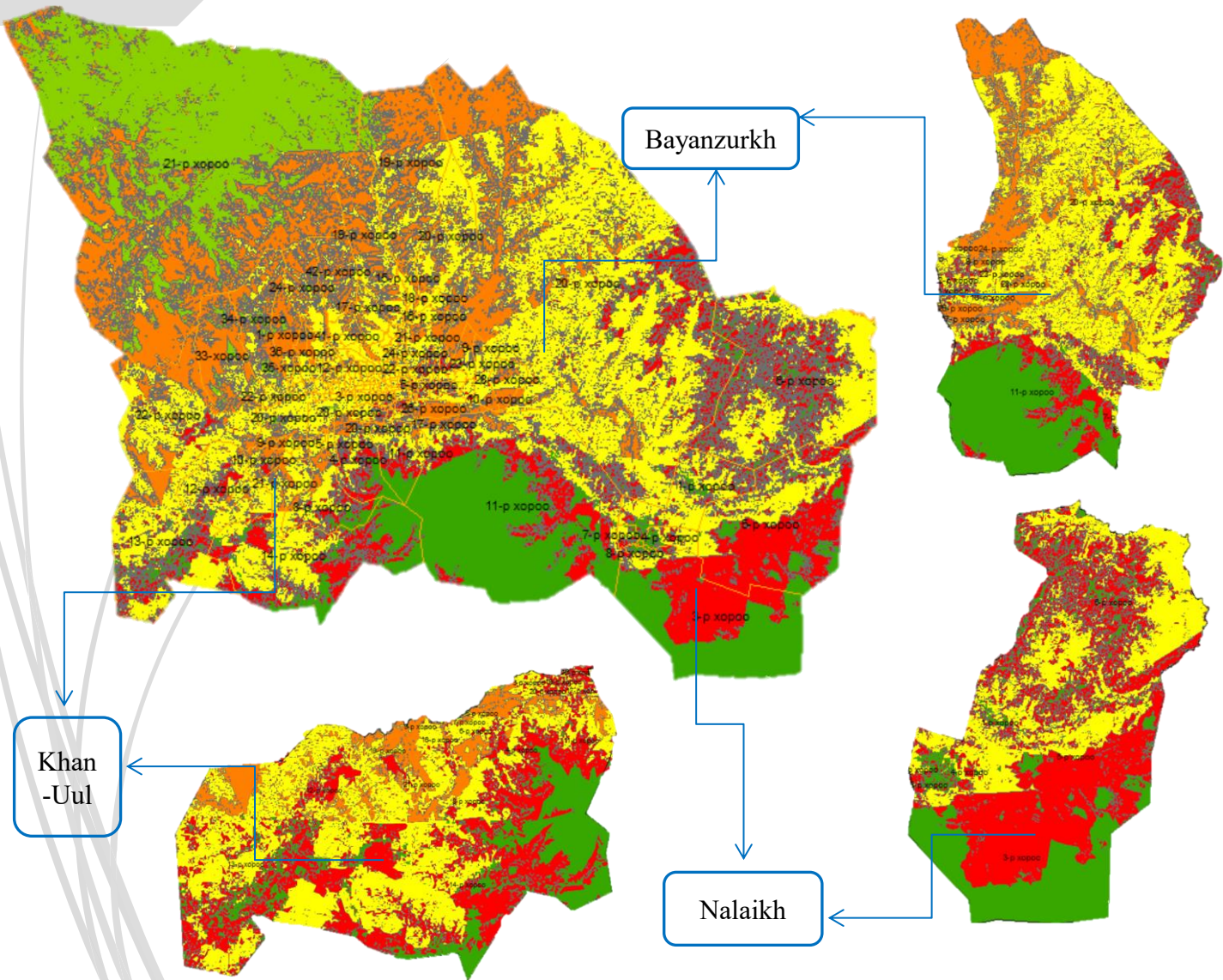


Mechanism: Topographic controls and steep-gradient runoff accumulation.

Risk: Alluvial fan morphology exacerbates flood propensity, funneling high-velocity water into urban fringes before it reaches the main basin.



DISTRICTS AND KHOROOS IN THE FLOOD RISK ZONES



Flood Risk	District	Khoroos	Area	Share (%)
high	Songinokhairkhan	Khoroos 9,21,26,33,8,35,25,34,36,40,42,1,24,10 are located in flood risk zones	1006.9 km ²	25.5
	Chingeltei	The entire territory of all khoroos is within the flood risk zone.		
	Sukhbaatar			
	Bayanzurkh			

very high	Khan-Uul	Khoroos 14,16,19,12,20,21,18,11,10,9,8,6,4,3 are located in high flood risk zones.	775.72 km ²	19.7
	Bayanzurkh	Khoroos 3,15,18,1,6,11 are located in high flood risk zones.		
	Nalaikh	Khoroos 6,5,3 are located in high flood risk zones.		



CONCLUSION

- The Analytic Hierarchy Process (AHP) was employed following a systematic evaluation of alternative flood risk assessment techniques, including hydrological modeling, hydraulic simulations, and rainfall-runoff analyses. This decision was driven by AHP's capacity to integrate and quantitatively compare heterogeneous geospatial datasets through pairwise criteria evaluation, enabling a robust multi-parametric assessment of flood susceptibility factors.
- Thematic layers were derived from ALOS PALSAR synthetic aperture radar (SAR) digital elevation models at 12.5-meter resolution, incorporating terrain parameters critical to flood modeling: elevation gradients, slope aspect, flow accumulation, and watershed boundaries. These processed layers serve dual purposes: as foundational inputs for hydrodynamic simulations and as standalone datasets for geomorphological studies, particularly in analyzing erosion patterns and land degradation susceptibility across variable topographies.
- The AHP derived flood risk matrix categorized the study area into five ordinal risk classes:
 - Very High-risk zones (775.7 km², 21.3% of study area)
 - High-risk zones (1006.89 km², 27.6%)
 - Moderate-risk zones (1218.7 km², 33.4%)
 - Low-risk zones (446.7 km², 12.2%)
 - No risk zones



ХӨДӨЛМӨРИЙН ГАВЬЯНЫ УЛААН ТҮГИЙН ОДОНТ
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*Thank you
for
your attention*



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